EV Market Performance and Insights

Comprehensive Analytical Project Report & Story Findings

> **Project Report:** Data Analytics with Tableau | **Prepared by:** Harsh Patidar | **Date:** 26 October 2026
> **Tableau Public Link:** [Visual Analytics Portfolio Link]

1. INTRODUCTION

1.1 Project Overview

The electric vehicle (EV) industry is one of the fastest-growing sectors of the global automotive market, driven by environmental awareness, government incentives, and rapid advances in battery technology. This project analyzes EV market data—encompassing sales volume, pricing dynamics, battery specifications, driving range, manufacturing countries, customer ratings, and annual CO2 savings—and presents actionable insights through two interactive Tableau dashboards and a dedicated Tableau Data Story.

1.2 Dashboard Architecture Summary

Dashboard 1 (Market Performance): Focuses on executive-level KPIs and competitive benchmarking. It includes overall market KPIs (Total EV Sales: 1.55B, Average Price: $64,921, Average Range: 324.54 km), global production output by country and vehicle category, top manufacturers rated by customer reviews vs. sales volume, driving range distributions across segments, and 3-year retained vehicle value.

Dashboard 2 (Technology and Impact): Focuses on deep-tech battery performance and environmental metrics. It tracks historical sales trends over model years, battery technology performance (sales units vs. cycle life across chemistries), charging power vs. energy efficiency, and regional environmental impact through CO2 abatement maps.

Tableau Data Story: Combines Dashboard 1 and Dashboard 2 into a guided interactive sequence, allowing decision-makers to transition seamlessly from macro market trends to granular battery and environmental analyses.

1.3 Purpose

The primary objective is to transform raw EV market datasets into intuitive, interactive visual stories. This empowers automotive OEMs, strategy consultants, and sustainability analysts to identify which vehicle categories drive adoption, which battery chemistries dominate, how charging speeds impact efficiency, and where environmental benefits are maximized.

2. IDEATION PHASE

2.1 Problem Statement

EV manufacturers and analysts generate massive volumes of sales, technical, manufacturing, and customer review data. Without a unified interactive visualization platform, monitoring core KPIs—such as annual sales growth, battery cycle degradation, driving range trade-offs, manufacturing geographical distribution, and customer satisfaction—remains highly challenging.

2.2 Persona Mapping

Persona	Role	Core Need & Focus
Emily Johnson	EV Market Strategy Director	Needs macro-level category sales trends, customer satisfaction scores, battery adoption rates, and range comparisons to identify high-growth segments for portfolio expansion.

Persona	Role	Core Need & Focus
David Chen	Automotive BI Manager	Requires manufacturer market share, sales volume distribution, production country breakdown, and resale value metrics to advise clients entering the EV ecosystem.
Dr. Sarah Williams	Sustainable Mobility Research Analyst	Analyzes battery cycle life, range improvements, energy efficiency vs. charging speed, and CO2 reduction to support evidence-based policy making.

2.3 Empathy Map Canvas

Quadrant	Emily (Strategy Director)	David (BI Manager)	Dr. Williams (Researcher)
Says	"Which vehicle segments offer optimal investment return?"	"Who holds the competitive market edge globally?"	"Which battery chemistry drives real adoption?"
Thinks	Needs rapid, executive-level summaries and reliable KPIs.	Clients demand hard data-backed proof, not opinions.	Policy frameworks require credible technical evidence.
Does	Reviews category sales, pricing, and satisfaction trends.	Benchmarks OEM performance, resale value, and regions.	Compares battery lifecycles, charging power, and CO2 impact.
Feels	Under pressure to optimize multi-million dollar portfolios.	Driven to deliver high-confidence client recommendations.	Motivated by long-term environmental sustainability goals.

3. DETAILED ANALYTICAL INSIGHTS & STORY ANALYSIS

3.1 Dashboard 1 Insights: Market Performance & Competitive Landscape

Global Executive KPIs: The global EV ecosystem exhibits total cumulative sales of 1.55 Billion units across all categories. The average vehicle selling price stands at $64,921 with a mean driving range of 324.54 km, resulting in 738,000 metric tons of annual CO2 savings.

Production Hub Dominance: Manufacturing output is heavily concentrated in Asia-Pacific and emerging assembly hubs, led by India, China, and Vietnam. These countries dominate volume manufacturing, particularly for two/three-wheelers and commercial EV fleets.

Driving Range Distribution & Vehicle Classes: Luxury Sedans and Premium SUVs achieve the highest single-charge range (averaging over 450 km), commanding top tier pricing ($80,000 - $120,000+). Conversely, Micro City Cars and Two-Wheeler Scooters occupy short-range brackets (80–180 km) at lower price points, serving urban last-mile mobility.

Resale Value Retention (3-Year): Commercial and high-utility EVs display the strongest 3-year retained asset values. Electric Buses lead with an average retained value of $14.5M across commercial fleets, followed by Electric Heavy Trucks ($11.2M), Sports Cars ($8.69M), and Luxury SUVs ($6.15M), reflecting strong commercial asset longevity.

OEM & Customer Satisfaction Benchmarking: High sales volume does not always directly correlate with top customer ratings. While dominant OEMs lead in raw sales units, specialized manufacturers with dedicated EV platforms achieve higher customer satisfaction scores due to superior software integration and build quality.

3.2 Dashboard 2 Insights: Technology, Efficiency & Environmental Impact

Sales Trend & Segment Growth: Annual EV sales maintain steady growth trajectories between 2018 and 2026. The highest volume growth is consistently driven by Two-Wheeler Scooters and Compact Urban EVs, which serve high-density metropolitan markets.

Battery Chemistry Performance: NMC (Nickel Manganese Cobalt) and LFP (Lithium Iron Phosphate) account for over 65% of total battery deployments. LFP and emerging Solid-State technologies offer the longest operational cycle life (exceeding 2,500–3,000 charge cycles) and lowest annual degradation, making them ideal for long-term ownership. NMC and NCA remain favored for high-density, long-range luxury vehicles.

Charging Power vs. Energy Efficiency: A clear engineering trade-off exists between charging speed and vehicle efficiency. High-capacity commercial vehicles and heavy SUVs support fast DC charging up to 350 kW but exhibit higher energy consumption per kilometer. Smaller passenger vehicles operate efficiently at 50–150 kW DC fast-charging rates.

Environmental & CO2 Impact: Geographic CO2 displacement map shows that regions replacing diesel commercial fleets and high-mileage urban commuting vehicles with EVs generate the highest annual CO2 savings per vehicle unit, directly accelerating global decarbonization targets.

3.3 Combined Story Arc Insights

By linking Dashboard 1 and Dashboard 2 into a Tableau Story, stakeholders can follow a complete narrative: from market size and OEM positioning, down to technical battery selections and localized environmental benefits. This integrated perspective confirms that mass-market adoption depends on expanding cost-effective LFP/Sodium-Ion platforms, while high-end profitability relies on energy-dense NMC and Solid-State architectures.

4. REQUIREMENT ANALYSIS & SOLUTION DESIGN

4.1 Requirement Matrix

Functional Requirement	Implementation in Tableau
Executive KPI Display	Single-card summary containers for Total Sales (1.55B), Avg Price ($64.9K), Avg Range (324.5 km), and CO2 Saved (738K tons).
Category & Regional Analysis	Global choropleth maps and stacked bar charts showing production distribution across India, China, Vietnam, US, and EU.
Competitive OEM Rating	Scatter plot and ranking bars plotting customer review scores against unit sales volume.
Battery Tech Comparison	Dual-axis bubble/bar charts comparing battery chemistry sales against cycle life and degradation rates.
Interactive Filtering	Global vehicle category color-legend filters cross-linking all charts across Dashboard 1 and Dashboard 2.

4.2 Technology Stack

Layer	Technology / Tool
Data Layer	EV Market Structured Dataset (CSV / Excel Format)
Analytics & Visualizations	Tableau Desktop (Calculated fields, parameters, LOD expressions)
Publishing & Distribution	Tableau Public Cloud Platform
Project Documentation	Microsoft Word (.docx) / Custom Analytics Report

5. FUNCTIONAL TESTING & VERIFICATION

Test Case Description	Expected Result	Status
KPI Card Accuracy	Displays Sales 1.55B, Price $64,921, Range 324.54 km, CO2 738K tons	**PASS**
Category Filter Action	Selecting 'Luxury SUV' dynamically filters all charts in Dashboard 1 & 2	**PASS**
Tooltip Hover Details	Hovering displays exact price, battery type, manufacturer, and rating	**PASS**
Tableau Public Rendering	Dashboards render fully with active interactions in web browser	**PASS**
Story Point Navigation	Tabbed story navigation switches seamlessly between visual chapters	**PASS**

6. STRATEGIC RECOMMENDATIONS & FUTURE SCOPE

6.1 Strategic Recommendations

- **For Strategy Directors:** Prioritize LFP/Sodium-Ion vehicle platforms for entry-level and high-volume urban segments to lower initial customer purchase costs while scaling Solid-State battery development for long-range premium models.
- **For BI & Market Managers:** Target underserved geographic markets by aligning product deployment with localized manufacturing hubs in India and Southeast Asia to minimize production logistics expenses.
- **For Policy & Sustainability Analysts:** Encourage policy incentives for fast-charging infrastructure (150–350 kW DC) along major transport corridors to maximize fleet CO2 abatement.

6.2 Future Scope

- **Live Data Pipeline:** Connect Tableau workbook to live cloud database feeds (e.g., Snowflake, AWS Redshift) for real-time sales and telematics tracking.
- **Predictive Analytics:** Incorporate Python/R integration in Tableau to forecast 5-year battery degradation and regional EV market share growth.
- **Charging Infrastructure Mapping:** Integrate geospatial layers showing public charging station density relative to EV sales growth.

7. APPENDIX & PROJECT LINKS

Author Name: Harsh Patidar

Project Topic: EV Market Performance and Insights Data Analytics

Tableau Public Demo Link: [Visualizations Portfolio URL]

GitHub Project Repository: [Repository URL]